· I/O
· TCP/IPs
7. Disk and memory usage optimisation
8. Information about graphics operations and performance
9. System logs
10. Boot process
11. Power consumption

The performance of different hardware varies based on the operating systems. But on the same hardware, the major Linux OSs like Ubuntu, Red Hat, Debian, Solaris, etc, perform differently because they have their own process execution techniques and methods. OS performance depends on internal architecture, system calls, context switching, file reading/creating/deleting, kernel compilation, system version, file system, etc.

System information

Figure 1 shows the processing parameters for system configuration, on which testing is done.

PHORONIX-TEST-SUITE.COM	Phoronix Test Suite 5.8.0
AMD C-60 APU @ 1.00GHz (2 Cores)	Processor
ASUS K53U v1.0	Motherboard
AMD Family 14h Root Complex	Chipset
1 x 4096 MB DDR3-667MHz	Memory
320GB Seagate ST9320325AS	Disk
ASUS AMD Radeon HD 6290 384MB (276MHz)	Graphics
AMD Wrestler HDMI Audio	Audio
Realtek RTL8111/8168/8411 + Qualcomm Atheros AR9285 Wireless	Network
Ubuntu 14.04	OS
3.16.0-30-generic (i686)	Kernel
Unity 7.2.4	Desktop
X Server 1.16.0	Display Server
radeon 7.4.0	Display Driver
GCC 4.8.2	Compiler
ext4	File-System
1366x768	Screen Resolution

Figure 1: System configuration

Parameters that reduce system performance

CPU frequency and number of cores

Processor families like Intel, AMD, SPARC, etc, have various process architectures with a range of frequencies. Processors with a good frequency and architecture give the best performance. Nowadays, processors have specific frequency units in GHz, which was not the case earlier. Earlier generations of processors had processing frequencies measured in MHz, so gaps in the performance of the CPU unit could be seen and analysed easily. CPU performance metrics are based on the following parameters.

i. **Response time**
- Processing speed
- Channel capacity
- Latency
- Bandwidth
- Throughput

ii. **Power consumption**
iii. **Environmental impact**

The amount of time taken by any processor or task can be termed as 'performance', which does not mean clock frequency alone or the number of instructions executed per clock cycle, but is the combination of clock frequency and instructions per clock cycle:

$P = I * F$

...where, P = performance, I = instructions executed per clock cycle and F= frequency

Performance testing on the cores of the CPU: A system has two cores, and when one core is being used and the other is disabled, the command for this process is:

```
nano /sys/devices/system/cpu/cpu1/online
```

Case 1, when both cores are enabled, is shown in Figure 2.

```
root@india-K53U:/home/india/Downloads# cat /proc/cpuinfo | grep processor
processor       : 0
processor       : 1
```

Figure 2: When both cores are enabled

The execution times of Gzip and Apache server are 455.71 and 73.78 seconds, respectively, as shown in Figures 4 and 6.

Case 2, when only one core is enabled, is shown in Figure 3.

```
root@india-K53U:/home/india/Downloads# cat /proc/cpuinfo | grep processor
processor       : 0
```

Figure 3: When only one core is enabled

The execution times of Gzip and Apache server are 847.36 and 175.55 seconds, respectively, as shown in Figures 5 and 7.

Apache build testing: This test shows how long it takes to build the Apache HTTP server.

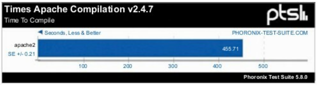

Figure 4: Compression of Apache when two cores are enabled

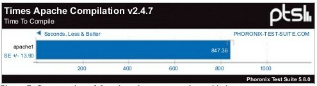

Figure 5: Compression of Apache when one core is enabled

Gzip compression testing: This test shows how long it takes to do Gzip compression of files.

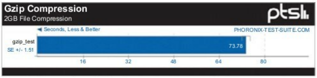

Figure 6: Compression of Gzip when two cores are enabled